

AUTOClave SCALDING

Lessons Learned

This Lessons Learned document focuses on an incident in which bottles of hot media in an autoclave burst, scalding a researcher. It demonstrates the importance of properly maintaining equipment, adequate handling practices, training, PPE use, and supervision of inexperienced researchers.

What Happened?

An undergraduate researcher was working alone in a lab. They placed twenty-two 500 mL bottles containing 440 mL each of bacterial media in a benchtop autoclave, planning to sterilize the media. They filled the autoclave tray with water, closed the autoclave door, and secured the door with a bungee cord. The bungee cord was used because the lab was having issues with the door not sealing well. The researcher started the sterilization cycle and left the lab.

About 15 minutes later, the researcher returned and found that the autoclave had aborted its cycle due to low water. The researcher opened the autoclave wearing heat-resistant gloves and lab-appropriate street attire but no other personal protective equipment (PPE). They went to the sink, filled a beaker with cool water, then began pouring the water into the autoclave tray. Some of the water splashed and hit the bottles. Two of the bottles shattered, spraying scalding media on the researcher's lower face and upper torso, causing first- and second-degree burns.

The researcher went to a nearby sink and rinsed the affected areas with cool water. They then returned to the autoclave to see if anything else was wrong. Around this time, they noticed the affected skin starting to peel and realized the burns might be serious. They called the lab manager, who advised them to call Environmental Health & Safety (EH&S). After reporting their injury by phone, they biked to an urgent care center. The urgent care center initially turned them away, as most urgent care centers do not treat workplace injuries, but because the researcher was an undergraduate they were eventually given medical care.

Why Did This Happen?

The immediate cause of the incident was the thermal shock to the bottles. When the cool water hit the still-hot bottles, it caused stress to the bottles' structure, leading to some breaking. The media in the bottles was hot, causing a small amount of outward pressure. This led to the liquid spraying out.

The root cause of this incident was the poor condition of the autoclave. The autoclave was approximately 20 years old, and had not received maintenance for at least three years. The door did not seal properly, and due to routine heavy loading, the autoclave shelf sagged. Additionally, the temperature gauge did not function. The most significant factor was the poor door seal. This issue had been present for at least three years. The poor seal frequently caused the autoclave to lose steam and thus run out of water mid-cycle, as happened in this incident. Lab members addressed this by using a bungee cord to more tightly hold the door handle

closed; when the elastic in the cord would break, the cord would be replaced. In this circumstance, the cord had failed several weeks prior to the incident but had not been replaced.

The exact cause of the poor seal is not known. The lab had not performed the daily cleaning recommended by the manufacturer, which may have caused material to build up on the seal surfaces. The door gasket may have been degraded and needed replacement. The screw that controls the tightness of the door's seal may have required adjustment. The ad hoc solution applied was, however, insufficient. Had the door sealed properly, the autoclave would not have run out of water, and the entire incident would have been averted.

Several other factors contributed to causing the incident and its severity. The lab's training and standard operating procedure (SOP) for using the autoclave did not cover a number of key safety practices required by [Stanford's Autoclave Safety Guidance](#). One critical omission was appropriate PPE (face shield and goggles, lab coat, rubber apron, heat resistant gloves). Because the lab was unaware of the appropriate PPE, they did not own a face shield or a rubber apron. Had the researcher been wearing the appropriate PPE, their injuries may have been prevented, or at least significantly reduced.

What Went Right?

The researcher promptly rinsed the affected skin with cool water, the recommended treatment for burns. They also promptly sought medical attention (although EH&S recommends calling 911 and/or proceeding directly to the Stanford Hospital Emergency Department in such circumstances). These factors helped mitigate the level of harm caused.

How to Prevent and Respond to Accidents Like This

- Perform manufacturer-recommended maintenance regularly. If a piece of equipment is malfunctioning, consult the owner's manual and/or contact the manufacturer for guidance. Only perform repairs recommended by the manufacturer. If you are uncertain about whether you will be able to safely conduct repairs, hire a professional. Do not attempt to use malfunctioning equipment.
- Develop thorough SOPs for hazardous laboratory procedures. Before writing, find and review relevant safety guidance, and ensure that the SOP contains established best practices. Ensure lab members are trained on the SOP before performing the procedure.
 - [EH&S has a number of SOP templates](#). You can find examples of well-written SOPs in the [SOP library](#). [You can submit SOPs for review by EH&S as well](#).
- Have the proper PPE available for all tasks performed in the lab. Ensure all lab members are trained on what PPE to wear. Principal Investigators (PIs) are responsible for ensuring lab members comply with lab PPE requirements.
 - Review your lab's [PPE Assessment Tool](#) as a group at least once per year.
- Have a more senior lab member supervise undergraduates when they are working. Labs should have policies covering what procedures undergraduates are allowed to perform and which are deemed too hazardous.
- Avoid working alone when possible. Discuss as a lab what procedures are too hazardous to be performed while working alone.
- Train all lab members to call 911 or go to the Stanford Hospital Emergency Department if they are seriously injured while at work.